Multiple Vertex Coverings by Cliques

Wayne Goddard*
Department of Computer Science
University of Natal
Durban, 4041 South Africa

Michael A. Henning†
Department of Mathematics
University of Natal
Private Bag X01, Scottsville, 3209 South Africa

Abstract

For positive integers $m_1, \ldots, m_k$, let $f(m_1, \ldots, m_k)$ be the minimum order of a graph whose edges can be coloured with k colours such that every vertex is in a clique of cardinality m_i all of whose edges have the i^{th} colour for all $i = 1, 2, \ldots, k$. The value for $k = 2$ was determined by Entringer et al. (*J. Graph Theory* **24** (1997), 21–23). We show that if k is fixed then $f(m, \ldots, m) = 2km - o(m)$. We also provide some exact values for $f(m, n, 2)$.

Keywords: k-edge-colouring, cliques, covers.

1 Introduction

Motivated by results on the so-called framing number [1], Entringer et al. [2] determined the minimum order of a graph where every vertex belongs to both a clique of cardinality m and an independent set of cardinality n. They called this minimum $f(m, n)$.

Theorem 1 (Entringer et al. [2]) *For $m, n \geq 1$,*

$$f(m,n) = \left\lceil (\sqrt{m-1} + \sqrt{n-1}\,)^2 \right\rceil = m + n - 2 + \left\lceil 2\sqrt{(m-1)(n-1)}\, \right\rceil .$$

*Current address: Dept of Computer Science, Clemson University, Clemson SC 29634, USA

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The result has been generalised by Füredi, Mubayi and West [3], and by Maharaj and the authors [5, 6, 7]. Galvin [4] gave another proof of Theorem 1 and observed that it could be viewed as a result about set systems.

In [5], Theorem 1 was interpreted as a result about two-colouring edges. In particular, $f(m, n)$ is the minimum order of a simple graph whose edges can be two-coloured with colours red and blue such that every vertex is in a clique of cardinality m all of whose edges are red and every vertex is in a clique of cardinality n all of whose edges are blue. This concept was then extended by replacing clique with an arbitrary graph, and by replacing two colours with multiple colours.

In this paper we consider the original problem of cliques but with three or more colours. For positive integers $m_1, m_2, \ldots, m_k$, let $f(m_1, m_2, \ldots, m_k)$ be the minimum order of a graph whose edges can be coloured with k colours, say colours $1, 2, \ldots, k$, such that every vertex is in a clique of cardinality m_i all of whose edges are coloured i for all $i = 1, 2, \ldots, k$. When $m = m_1 = m_2 = \cdots = m_k$, we denote $f(m_1, m_2, \ldots, m_k)$ by $f_k(K_m)$. For example, as a special case of Theorem 1 we have:

Fact 2 *For $m \geq 2$, $f_2(K_m) = 4(m-1)$.*

Note the obvious:

Observation 3 *For $k \geq 3$, $f(m_1, m_2, \ldots, m_k) \geq f(m_1, \ldots, m_{k-1})$.*

2 Equal Cliques

In this section we establish bounds on $f_k(K_m)$.

For $k = 3$ and small m, the following results were established in [5].

Fact 4 ([5])
(a) $f_3(K_2) = 4$.
(b) $f_3(K_3) = 9$.
(c) $f_3(K_4) \leq 16$.

It was also noted in [5] that the question of $f_k(K_m)$ is connected to packings and designs, at least for m small relative to k. For example, for $k \equiv 1 \pmod 3$, since Kirkman triple systems of order $2k+1$ exist, it holds that $f_k(K_3) = 2k+1$. As another example, the complete graph K_{16} is edge-decomposable into five copies of $4K_4$, so that $f_5(K_4) = 16$.

We can show:

Theorem 5 $15 \leq f_3(K_4) \leq 16$.

Proof. The lower bound follows from the fact that there is no 3-colouring of the edges of K_{14} such that every vertex is in a 4-clique of each colour. We have only a computer proof. $\square$

In general, there is an upper bound from [5].

Lemma 6 ([5]) *For $m \geq 2$, $f_k(K_m) \leq 2k(m-1)$.*

For $k = 3$ this specialises to:

Fact 7 *For $m \geq 2$, $f_3(K_m) \leq 6(m-1)$.*

We conjecture that the upper bound is the correct value for all m sufficiently large. We show next that for fixed k it holds that $f_k(K_m) \geq 2km - o(m)$.

We will need the solution to a simple optimisation problem.

Lemma 8 *Fix some integers $i \geq 1$ and $k \geq 3$. For $1 \leq r \leq k$ and $1 \leq t \leq i$, let α_r^t be nonnegative real numbers such that $\alpha_r^t + \alpha_s^t \geq 1$ for all r, s and t with $r \neq s$.*

For $1 \leq r \leq k$, define $A_r^0 = 1$ and suppose $A_r^t \geq 1 + A_r^{t-1}\alpha_r^t$ for $1 \leq t \leq i$. Then

$$\sum_{r=1}^{k} A_r^i \geq k(2 - 2^{-i}).$$

Proof. Clearly we may assume that $A_r^t = 1 + A_r^{t-1}\alpha_r^t$ for all t and r.

Let $y^t = \min_r \alpha_r^t$, say $y^t = \alpha_1^t$. Then $\alpha_r^t \geq 1 - y^t$ for all $r \neq 1$. It follows that the vector $\underline{\alpha}^t = (\alpha_r^t)_{r=1}^k$ majorises $(y^t, 1-y^t, 1-y^t, \ldots, 1-y^t)$. If $y^t \geq \frac{1}{2}$, then it majorises $\underline{\frac{1}{2}} = (\frac{1}{2}, \frac{1}{2}, \ldots, \frac{1}{2})$. So in fact, since we are proving a lower bound, we may assume the vector $\underline{\alpha}^t$ is some permutation of $(y^t, 1-y^t, 1-y^t, \ldots, 1-y^t)$ for some $0 \leq y^t \leq \frac{1}{2}$.

If we fix the vectors $\underline{\alpha}^\tau$ for $\tau \neq t$, then the sum $\sum_{r=1}^k A_r^i$ is a linear function of y^t. So it is minimised at an extremal value; i.e., $y^t = 0$ or $\frac{1}{2}$. Thus we may assume $\underline{\alpha}^t$ is $\underline{\frac{1}{2}}$ or some permutation of $(0, 1, 1, \ldots, 1)$.

Now define for $1 \leq r \leq k$ and $0 \leq j < i$

$$E_r^i(j) = \prod_{t=j+1}^{i} \alpha_r^t,$$

and $E_r^i(i) = 1$. Observe that

$$A_r^i = \sum_{j=0}^{i} E_r^i(j).$$

Now, suppose $\underline{\alpha}^t = \underline{\frac{1}{2}}$ for all $t \leq i$. Then $E_r^i(j) = 2^{j-i}$, and $\sum_{r=1}^k A_r^i = k(2 - 2^{-i})$.

Otherwise, let m be the largest value such that $\underline{\alpha}^m \neq \underline{\frac{1}{2}}$. Then for $j \geq m$ we have $E^i_r(j) = 2^{j-i}$ for all r. Also, $E^i_r(m-1) = 2^{m-i}$ for $k-1$ values of r (and is zero for the other value of r). Thus,

$$\sum_{r=1}^{k} \sum_{j=m-1}^{i} E^i_r(j) = 2k - 2^{m-i}.$$

If $m = 1$, the result follows (using $k \geq 2$).

So, assume $m > 1$. Then either $E^i_r(m-2) = 2^{m-i}$ for (at least) $k-2$ values of r, or $E^i_r(m-2) = 2^{m-i-1}$ for $k-1$ values of r. In any event, $\sum_{r=1}^{k} E^i_r(m-2) \geq 2^{m-i}$, where we use that $k \geq 3$. It follows that $\sum_{r=1}^{k} A^i_r \geq 2k$. $\Box$

Theorem 9 *For fixed k, $f_k(K_m) \geq 2km - O(\log m)$.*

Proof. Let H be a graph with the edges k-coloured such that every vertex is in a clique of order m in each colour. Say the colours are $C_1, \ldots, C_k$.

Consider any vertex. It is in an m-clique in each colour. So we can find k disjoint cliques B^0_r of cardinality $m-1$ with B^0_r of colour C_r for $1 \leq r \leq k$. We will define a sequence of vectors of sets (B^i_r) such that for each i the k sets B^i_r are disjoint, and B^i_r is the union of $i+1$ cliques of colour C_r for $1 \leq r \leq k$. We will denote their cardinalities by $b^i_r = |B^i_r|$.

For $r, s \in \{1, \ldots, k\}$ with $r \neq s$, let δ^1_{rs} denote the minimum number of edges of colour C_s by which a vertex of B^0_r is joined to B^0_s. Then since an edge has only one colour, we have the bound:

$$b^0_r \, \delta^1_{rs} + b^0_s \, \delta^1_{sr} \leq b^0_r \, b^0_s.$$

Now let

$$e^1_{rs} = \delta^1_{rs} / b^0_s.$$

It follows that $e^1_{rs} + e^1_{sr} \leq 1$.

Now, let v be the vertex of B^0_s for which δ^1_{sr} is attained and consider the m-clique of colour C_r containing v. Apart from possibly a vertex of each B^0_t, $t \neq r, s$, the remaining vertices of the clique must be in B^0_r or $V - \bigcup_t B^0_t$. That is, there is a clique of colour C_r, disjoint from $\bigcup_t B^0_t$ of cardinality at least $m - k + 1 - \delta^1_{rs}$.

Now, define B^1_r by adding to B^0_r a clique of colour C_r, disjoint from $\bigcup_t B^0_t$, of maximum cardinality. Then, ensure that the B^1_r are disjoint: this can be achieved by removing at most $k-1$ vertices from each set. Thus each B^1_r is the union of at most two cliques of colour C_r, and

$$b^1_r \geq b^0_r + m - 2(k-1) - \min_{s \neq r} \delta^1_{sr}.$$

Now repeat the argument using the B^1_r. Again it follows that $b^1_r \delta^2_{rs} + b^1_s \delta^2_{sr} \leq b^1_r b^1_s$ and that

$$e^2_{rs} + e^2_{sr} \leq 1.$$

The only difference is that the C_s-coloured m-clique for a vertex in B_r^1 can in theory contain one other vertex of B_r^1 and two vertices from each B_t^1 for $t \neq r, s$.

It follows that we can extend the collection B_r^1 of k disjoint sets to a collection B_r^2 of k disjoint sets, each consisting of three like-coloured cliques, such that $b_r^2 \geq b_r^1 + m - 3(k-1) - \min_{s\neq r} \delta_{sr}^2$. Continuing in this way, it follows that for $i \geq 1$,

$$b_r^i \geq b_r^{i-1} + m - (i+1)(k-1) - \min_{s\neq r} \delta_{sr}^i.$$

That is,

$$b_r^i \geq m - (i+1)(k-1) + b_r^{i-1}(1 - \min_{s\neq r} e_{sr}^i).$$

Now define $A_r^t = b_r^t/(m-(i+1)(k-1))$. Thus for $1 \leq t \leq i$

$$A_r^t \geq 1 + A_r^{t-1}(1 - \min_{s\neq r} e_{sr}^t).$$

Define

$$\alpha_r^i = 1 - \min_{s\neq r} e_{sr}^i.$$

Since all $e_{sr}^i \leq 1$, it follows that $\alpha_r^i \geq 0$. Further,

$$\alpha_r^t + \alpha_s^t \geq (1 - e_{sr}^t) + (1 - e_{rs}^t) \geq 1,$$

since $e_{sr}^t + e_{rs}^t \leq 1$.

Hence the A_r^t and α_r^t obey the hypotheses of Lemma 8. So by that lemma, $\sum_{r=1}^k A_r^i \geq k(2-2^{-i})$. Thus

$$n \geq \sum_{r=1}^{k} b_r^i \geq k(2-2^{-i})\,(m-(i+1)(k-1)) \geq 2km - k2^{-i}m - 2(i+1)k^2.$$

Now, set $i = \log_2 m$ and conclude. □

We note that the above technique can provide a lower bound in general for $f(\ell, m, n)$ (though we have not determined the form). However, this bound does not even asymptotically match the upper bounds given in the following section. For example, the best upper bound for $f((1+\varepsilon)m, m, (1-\varepsilon)m)$ we have is $6m - O(1)$, but the lower bound does not have the same coefficient for m.

3 Three Cliques

3.1 General upper bounds

In this section, we establish two upper bounds on $f(\ell, m, n)$. The first extends a construction given in [3, 5].

Theorem 10 *For integers ℓ, m, n with $\ell, m, n \geq 2$, it holds that $f(\ell, m, n) \leq 2(\ell+m+n-3)$.*

Proof. We 3-colour the edges of $G = K_{2(\ell+m+n-3)}$ as follows. The vertex set is partitioned into six sets L_1, L_2, M_1, M_2, N_1 and N_2 of cardinalities $\ell-1$, $\ell-1$, $m-1$, $m-1$, $n-1$ and $n-1$, respectively. The edges in L_1 and L_2 are coloured lilac, those in M_1 and M_2 magenta, and those in N_1 and N_2 navy. For $i = 1, 2$, we colour all edges between L_i and M_i magenta and all edges between L_i and M_{3-i} lilac. Furthermore, for $i = 1, 2$, we colour all edges between $L_i \cup M_i$ and N_i navy, all edges between L_i and N_{3-i} lilac, and all edges between M_i and N_{3-i} magenta. The remaining edges (those joining L_1 and L_2 or M_1 and M_2 or N_1 and N_2) are coloured arbitrarily (or discarded).

We have explicitly placed each vertex in a lilac clique of cardinality ℓ, a magenta clique of cardinality m, and a navy clique of cardinality n. Thus $f(\ell, m, m) \leq 2(\ell-1+m-1+n-1)$. $\square$

When $n = 2$, the result in Theorem 10 can be improved as noted in Theorem 14. Next we establish an upper bound on $f(\ell, m, n)$ when ℓ is at least the sum of m and n. This generalises the construction from [2].

Theorem 11 *For integers ℓ, m, n with $\ell \geq m+n-1$ and $m, n \geq 2$, it holds that $f(\ell, m, n) \leq \ell+m+n-2+2\lceil\sqrt{(\ell-1)(m+n-2)}\,\rceil$.*

Proof. Let $\alpha = \sqrt{(\ell-1)(m+n-2)}$. Set $x = \ell-1+\lceil\alpha\rceil$, $y = m-1+\lceil(\ell-1)(m-1)/\alpha\rceil$, and $z = n-1+\lceil(\ell-1)(n-1)/\alpha\rceil$.

We 3-colour the edges of $G = K_{x+y+z}$ as follows. The vertex set is partitioned into three sets X, Y and Z of cardinalities x, y and z, respectively. The edges in X are coloured lilac, those in Y magenta, and those in Z navy.

Now, for each vertex in Y choose $\ell-1$ edges joining it to X such that the ends in X are distributed as evenly as possible and colour these lilac. (For instance, if $Y = \{v_1, \ldots, v_s\}$ and $X = \{u_0, \ldots, u_{x-1}\}$ then for $1 \leq i \leq s$ join v_i to the vertices u_j for $(\ell-1)(i-1) \leq j \leq (\ell-1)i-1$, where subscripts are modulo x.) Colour the remaining edges between Y and X magenta.

Next, for each vertex in Z choose $\ell-1$ edges joining it to X such that the ends in X are distributed as evenly as possible and colour these lilac. Colour the remaining edges between Z and X navy. Finally, for each vertex in Z choose $m-1$ edges joining it to Y such that the ends in Y are distributed as evenly as possible and colour these magenta. Colour the remaining edges between Z and Y navy.

We have explicitly placed each vertex of Y and Z in a lilac clique of cardinality ℓ and these cliques cover X. Furthermore, we have explicitly placed each vertex of Z in a magenta clique of cardinality m and these cliques cover Y.

Each vertex of X is joined to at most $\lceil y(\ell-1)/x\rceil$ vertices in Y with lilac edges. Thus each vertex of X is joined by at least $y - \lceil y(\ell-1)/x\rceil$ magenta edges to vertices in

Y, and so is in a magenta clique of cardinality $y - \lceil y(\ell-1)/x \rceil + 1$. We need to show $y - \lceil y(\ell-1)/x \rceil + 1 \geq m$. If we substitute for x and y and simplify, that inequality is equivalent to

$$\left\lceil \frac{(\ell-1)(m-1)}{\alpha} \right\rceil \cdot \lceil \alpha \rceil \geq (\ell-1)(m-1),$$

which is true. Thus, each vertex of X is in a magenta clique of cardinality m and these cliques cover Y. Similarly, each vertex of X is in a navy clique of cardinality n, and these cliques cover Z.

Finally, each vertex of y is joined to at most $\lceil z(m-1)/y \rceil$ vertices in Z with magenta. Thus each vertex of Y is joined by at least $z - \lceil z(m-1)/y \rceil$ navy edges to vertices in Z, and thus is in a navy clique of cardinality $z - \lceil z(m-1)/y \rceil + 1$. We need to show that $z - \lceil z(m-1)/y \rceil + 1 \geq n$. On substituting for y and z and simplifying, that inequality is equivalent to

$$\left\lceil \frac{(\ell-1)(n-1)}{\alpha} \right\rceil \cdot \left\lceil \frac{(\ell-1)(m-1)}{\alpha} \right\rceil \geq (m-1)(n-1).$$

This is implied by $\ell - 1 \geq \alpha$. Since by hypothesis $\ell - 1 \geq m + n - 2$, this is true. Thus, each vertex of Y is in a navy clique of cardinality n and these cliques cover Z.

We have therefore shown that every vertex of G is in a lilac clique of cardinality ℓ, a magenta clique of cardinality m, and a navy clique of cardinality n. Thus, $f(\ell, m, n) \leq x + y + z$, and so

$$f(\ell, m, n) \leq \ell + m + n - 3 + \lceil \alpha \rceil + \left\lceil \frac{(\ell-1)(m-1)}{\alpha} \right\rceil + \left\lceil \frac{(\ell-1)(n-1)}{\alpha} \right\rceil. \qquad (1)$$

Since $(\ell-1)(m-1)/\alpha + (\ell-1)(n-1)/\alpha = \alpha$ and in general $\lceil a \rceil + \lceil b \rceil \leq \lceil a+b \rceil + 1$, it follows that $f(\ell, m, n) \leq \ell + m + n - 3 + 2\lceil \alpha \rceil + 1$, as required. □

It seems possible that the bound of $f(\ell, m, n)$ given in Inequality (1) is the correct value for $\ell \gg m, n \gg 1$.

3.2 Tiny Cliques

In this final section, we consider the case where one or more of the cliques has size 2. We look first at two small cliques of size 2; that is, $f(m, 2, 2)$.

An m-clique is a complete subgraph of order m. We say that a graph is *m-clique-minimal* if every vertex is in an m-clique, but the removal of any edge leaves some vertex not in an m-clique. We shall need the following result from [2].

Lemma 12 (Entringer et al. [2]) *Let G be m-clique-minimal with U the set of vertices of degree $m-1$. Then for every edge e there is a vertex $u \in U$ whose unique m-clique contains e.*

Theorem 13 *For $m \geq 3$, $f(m,2,2) = m + \left\lceil \sqrt{8(m-2)} \right\rceil$.*

Proof. **(a)** We start by proving the lower bound. Let G be an m-clique-minimal graph on p vertices such that every vertex has at least two nonneighbours. We will show that $p \geq m + \sqrt{8(m-2)}$.

Consider the set U of vertices of degree $m-1$ in G. Since the neighbourhood of a vertex in U is complete, the subgraph induced by U is the disjoint union of cliques, say $C_1, \ldots, C_t$. Each of these C_i extends to a clique of cardinality m, say D_i. By Lemma 12, every edge of G is in some D_i.

Now construct U' by taking from each clique C_i two representatives if C_i has cardinality at least 2, otherwise the unique vertex.

Claim: Every vertex x not in U' has at least two nonneighbours in U'.

Proof. Consider first a vertex $x \in V - U$. Note that since vertices in U have complete neighbourhoods, a vertex is either adjacent to all of a particular C_i or to none of it. Assume first that all of x's nonneighbours are in U. Then it has two nonneighbours in U. If these nonneighbours occur in two or more of the C_i, then x is nonadjacent to at least two vertices in U'. If x is adjacent to all but one clique, say C_i, then that clique cannot be a singleton and so two representatives of it are in U'. So in any event U' contains two nonneighbours of x.

Otherwise, x has a nonneighbour y outside U. Since y has degree more than $m-1$, it is in at least two of the D_i, and hence there are at least two vertices of U' to which it is adjacent. Neither of these two vertices can be adjacent to x.

Consider secondly a vertex $x \in U - U'$. Its neighbours are confined to one clique C_i. Then the result follows unless $t = 1$, or $t = 2$ and one of the cliques is a singleton. But that means G is the edge-sum of at most two cliques, and since every vertex has a nonneighbour, these cliques are disjoint and the claim holds. This proves the claim. □

Suppose $|U'| = s$. The number of edges leaving U' is at least $s(m-2)$. By the claim, the number of non-edges entering U' is at least $2(p-s)$. Thus

$$s(m-2) + 2(p-s) \leq (p-s)s,$$

or, equivalently, $s^2 - (p+4-m)s + 2p \leq 0$. This inequality has a real solution for p if and only if the discriminant $(p+4-m)^2 - 8p$ of the quadratic is nonnegative. This holds if and only if $p \geq m + \sqrt{8(m-2)}$ (since $p \geq m+2$).

(b) We now prove the upper bound by constructing a suitable graph. For $m = 3$ take two disjoint red triangles and add both a blue and a green matching between the cliques. For $m \geq 4$ proceed as follows.

Construct $G_p(a)$ on p vertices with $2 \leq a \leq p/4$ as follows. Start with two red cliques A and B on $2a$ and $p-2a$ vertices. Partition A into a pairs $A_1, \ldots, A_a$, and partition B

into a blocks, $B_1,\ldots,B_a$, each of which has either $\lfloor p/a\rfloor-2$ or $\lceil p/a\rceil-2$ vertices. Then, for $1\le i\le a$, A_i has two vertices and B_i has at least two vertices. For $1\le i\le a$, add blue and green edges between A_i and B_i such that each vertex in $A_i\cup B_i$ is incident with both a blue and a green edge. Finally, add all red edges between A_i and B_j for all $i\ne j$.

Note that every vertex is incident with a blue edge and a green edge. Every vertex of B is in a red clique of cardinality $p-2a$ and every vertex of A is in a red clique of cardinality

$$\alpha=p-2a-(\lceil p/a\rceil-2)+2.$$

Thus every vertex is in a red clique of this cardinality.

Now take $G_p(a)$ with $p=m+\lceil\sqrt{8(m-2)}\,\rceil$ and $a=[\sqrt{(m-2)/2}\,]+1$ where $[x]$ denotes the nearest integer to x. It is easily checked that $2\le a\le p/4$. It remains to show that for these values that $\alpha\ge m$.

By integrality, it suffices to show that $p-2a-p/a+4\ge m$ or that

$$ap-2a^2-p+4a-ma\ge 0,$$

having multiplied through by a. If we substitute the above values of a and p and set $m=\mu+2$, the left-hand side of the above inequality is equivalent to

$$-\mu+\left(\lceil\sqrt{8\mu}\,\rceil-2[\sqrt{\mu/2}\,]\right)[\sqrt{\mu/2}\,].$$

If we set $\mu=2s^2+t$, for s,t integral and $-2s+1\le t\le 2s$, then $[\sqrt{\mu/2}\,]=s$ and $\lceil\sqrt{8\mu}\,\rceil=4s+\lceil t/s\rceil$ for this range. So the above expression simplifies to

$$s\left\lceil\frac{t}{s}\right\rceil-t$$

which is indeed nonnegative. □

Note that Theorem 11 is applicable provided $\ell\ne m$:

$$f(\ell,m,2)\le\ell+m+2\left\lceil\sqrt{(\ell-1)m}\right\rceil,$$

while

$$f(\ell,m,2)\ge f(\ell,m)=\ell+m-2+\left\lceil 2\sqrt{(\ell-1)(m-1)}\right\rceil.$$

If the two large cliques are the same size, the lower bound gives the exact value, as shown by the following general upper bound which improves on Theorem 10.

Theorem 14 *For integers ℓ,m,n with $\ell,m\ge 2$, it holds that $f(\ell,m,2)\le 2(\ell+m-2)$.*

Proof. Consider the complete graph $G=K_{2(\ell+m-2)}$. Let A,B,C,D be a partition of the vertex set into four subsets, with A, B of cardinality $\ell-1$ and C, D of cardinality $m-1$.

Colour blue all edges between A and C or between B and D or with both ends in A or both ends in B. Colour red all edges between A and D or between B and C or with both ends in C or both ends in D. Colour the edges between A and B or between C and D green. Then each vertex is in a blue clique of cardinality ℓ, a red clique of cardinality m and a green clique of cardinality 2. Hence $f(\ell, m, 2) \leq 2(\ell + m - 2)$. $\square$

Corollary 15 *For $\ell \geq 2$ and $m > \ell - 2\sqrt{\ell - 1} + 1$, $f(\ell, m, 2) = f(\ell, m) = 2(\ell + m - 2)$.*

Proof. For this range of ℓ and m, it holds that $\left\lceil 2\sqrt{(\ell-1)(m-1)} \right\rceil = (\ell + m - 2)$. Thus $f(\ell, m, 2) \geq f(\ell, m) = 2(\ell + m - 2)$. $\square$

In the case that $m = \ell$, it follows that $f(m, m, 2) = 4(m-1)$, which generalises Fact 4(a). In fact, the construction easily extends to show that $f(m, m, 2, \ldots, 2) = f(m, m) = 4(m-1)$ provided there are at most $m - 1$ 2's.

We do not have a general result for $f(\ell, m, 2)$. The first case that is not handled by the above results is $f(7, 3, 2)$ which is at least 15 and at most 16.

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